



Back-end development

Objectives

- ◆ What is Front-End, Back-End, FullStack?
- ◆ Understand the basic of Data Structures and Algorithms.
- ◆ Understand the basic of Programming language and Framework.
- ◆ What is Web Hosting?
- ◆ What is CI/CD?

What is Front-End Development

- ♦ Front End development and Front End programming languages create what a user interacts.
- ♦ HTML, CSS, and JavaScript are the three Front End languages
- ♦ Each language has a specific function.
- ♦ Developers use frameworks and code libraries for efficiency as Bootstrap, JQuery,...



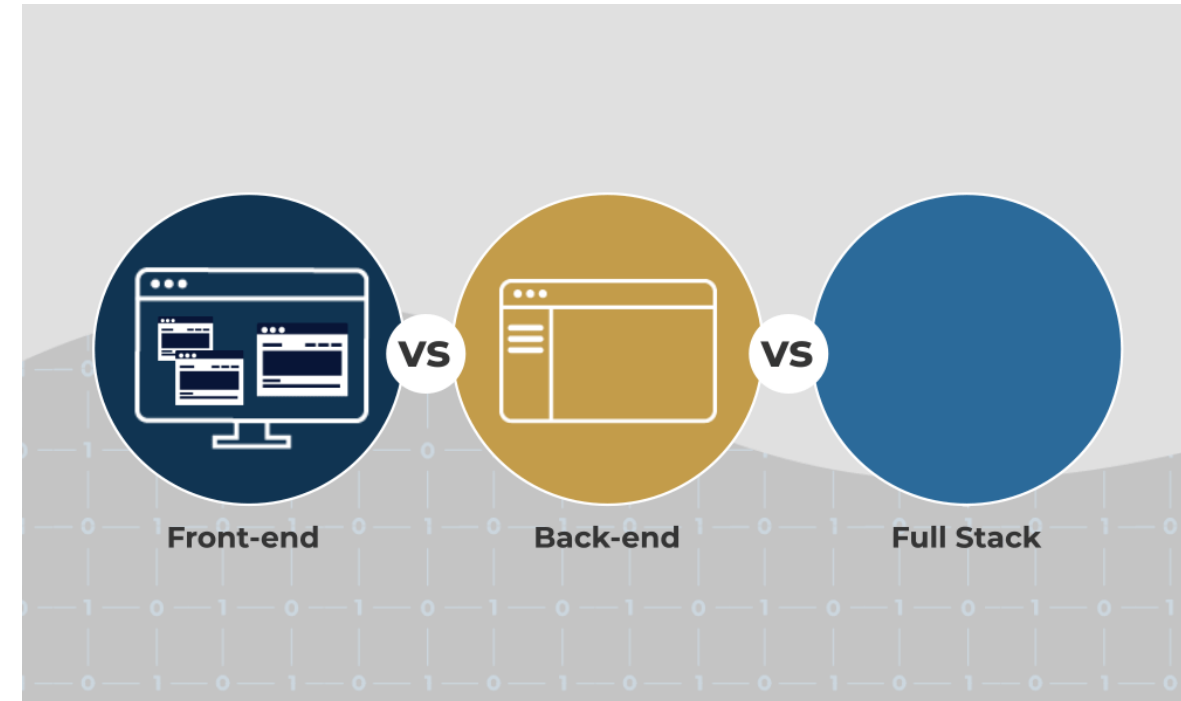
What is Back-End Development

- A Back End dev uses a set of Back End developer languages to bring the Front End language of a developer to life.
- Back End dev is creating the brains and logic of the website.
- Back End languages retrieve information from a database stored in a server that's then processed through an application



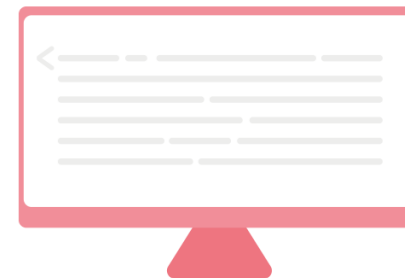
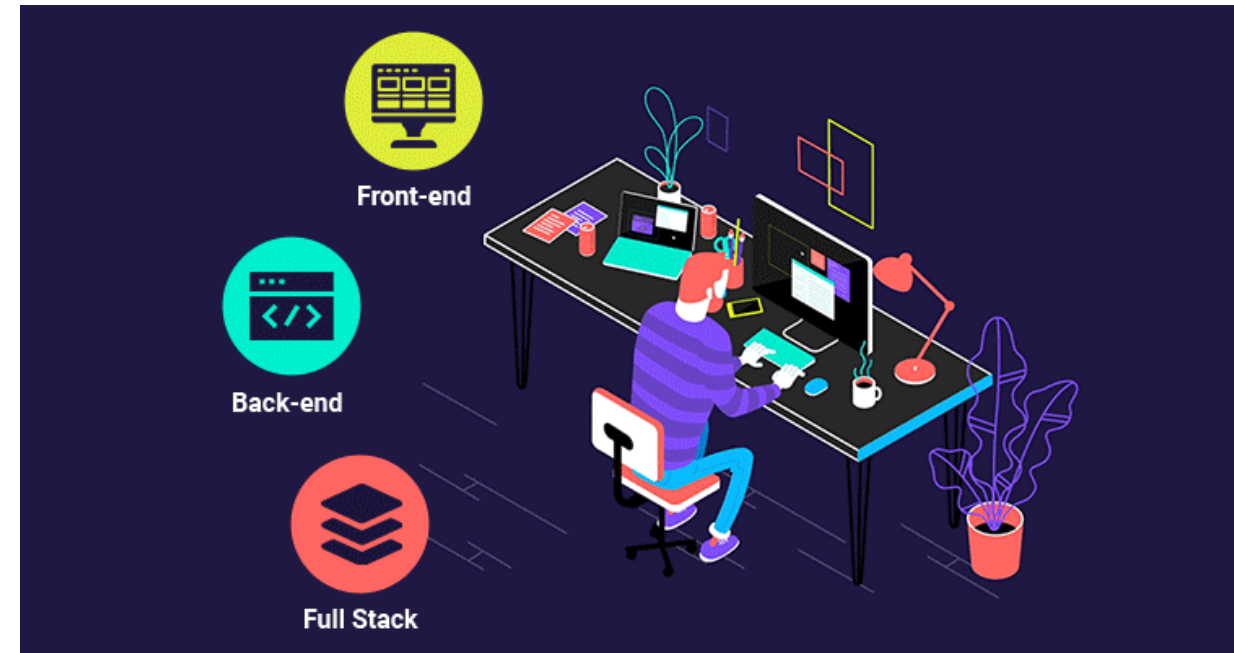
What is Full Stack?

- Full Stack development covers the full range of web development
- Full Stack devs are not experts in both Front End and Back End development
- Startups and smaller companies have popularized the role



Front End vs. Back End vs. Full Stack

- Three very popular developer roles
- Front End and Back End are specialized while Full Stack is generalized
- Full Stack uses Front End and Back End programming languages



Back End Developer Skills

- Array of skills are needed to be successful as a backend developer:
 - Python
 - Java
 - PHP
 - SQL
 - Git
 - HTML
 - CSS
 - JavaScript
 - Communication

Back-End Tool and Technology

- Servers

- The server is the engine of the network out of the four parts of your back-end stack, whether it's on-premises or in the cloud.
- These powerful computers provide the shared resources such as file storage, security and encryption, databases, email, and web services that networks require to function.

- Databases

- Databases serve as the brains that give websites their dynamic nature
- SQL vs. NoSQL
- How to choose the right database for your web application

Data Structures and Algorithms

- A backend developer creates code that does relational mapping to retrieve data from a database.
- MySQL, PostgreSQL, SQL SERVER, MongoDB, and Oracle Database are some of the most widely used DBMS. It's crucial to have a good understanding of how they work.



Stacks



Queues



Trees and Graphs



Sorting



Searching

Programming Languages and Frameworks

- Programming languages and frameworks are next on the list.
- An in-depth understanding of backend programming languages is one of the most important skills of a backend developer. This comprises the items listed below.



Programming Languages and Frameworks – cont'd

- **NodeJS** and **ExpressJS** are examples of JavaScript environments.
- **Java**, in particular, was designed from the start to be used on the server. Spring and Java Server Faces are two popular Java frameworks.
- **Python** is the most widely used programming language. It's adaptable and simple to use. Backend development is done with Python frameworks like Django and Flask.
- **C# language** is the recommended architecture for backend programming.
- **PHP, Perl, and Ruby** are some of the other languages available. Because backend and frontend developers collaborate so closely, having a basic understanding of front-end web technologies like HTML and CSS is also beneficial.

Version Control

- Version control systems aid in the management of project files and the tracking of the user's whole history. Git, GitHub, and Subversion are some of the most popular ones.
- Git aids in the management of project files. It maintains track of everything the user has worked on in the past. GitHub is a web-based version control service using Git.
- The user can examine other people's code, spot errors, and even suggest modifications.



Web Hosting Platforms

- Web hosting solutions enable you to place your product on a cloud service provider and access it via the internet.
 - Amazon Web Services(AWS)
 - Google Cloud Platform(GCP)
 - Microsoft Azure.
 - Heroku.
 - Firebase.



Back-End Developer Tasks

- Back-end developers are required to have technical expertise, analytical thinking, and excellent collaboration skills. Here's what many back-end developers do on a day-to-day basis:

Build and
maintain
websites

Perform quality
assurance
(QA) testing

Troubleshoot
and debug

Write high-
quality code

Assess
efficiency and
speed

Train and
support

Back End Developer Responsibilities

- To design effective and efficient solutions
- Application Programming Interfaces (APIs) development and administration.
- Develop data acceptance and storage solutions for websites, particularly for those involved in payment processing.
- Write, test, and maintain development solutions for code-related problems are all part of the job.
- To identify new features, communicate effectively with developers, designers, and system administrators.
- Create a website architecture by utilizing correct product lifecycle approaches, such as Agile Scrum and frameworks.
- Organize the system logic.
- Provide remedies to difficulties with the system.
- Debug and troubleshoot apps.

Common skills needed for Back-End Development

- Having a fundamental understanding of HTML, JavaScript, and CSS.
- Practical experience with back-end development (configuration, administration, and management of databases and servers).
- Languages used on servers, such as JavaScript (Node.js), PHP, C#, and Java.
- Database engineering (e.g., MySQL and SQL).
- Best practices for security and authentication. Proficiency in creating RESTful APIs

What is CI/CD?

- CI/CD, which stands for continuous integration and continuous delivery/deployment, aims to streamline and accelerate the software development lifecycle.
- Continuous integration (CI)** refers to the practice of automatically and frequently integrating code changes into a shared source code repository.
- Continuous delivery and/or deployment (CD)** is a 2 part process that refers to the integration, testing, and delivery of code changes.



Summary

- ◆ Concepts were introduced:
 - ◆ What is Front-End?
 - ◆ What is Back-End?
 - ◆ What is FullStack?
 - ◆ Understand the basic of Data Structures and Algorithms.
 - ◆ Understand the basic of Programming language and Framework.
 - ◆ What is Web Hosting?
 - ◆ What is CI/CD?